YUEWEN HOU

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OBJECTIVE

A second-year CS Ph.D. student seeking quantum computing research and software development internships. My research focuses on **classical–quantum hybrid system design**, **hardware-informed quantum circuit compilation**, **quantum error correction**, and **high-performance classical simulation for quantum systems**.

EDUCATION

Ph.D. in Computer Science, University of Michigan, Ann Arbor 2024–Present

Overall GPA: 4.00/4.00, Advisor Gokul Ravi

Bachelor of Engineering in Computer Science, University of Michigan, Ann Arbor 2022–2024

Advisors: Gokul Ravi, Baris Kasikci, Brendan Kochunas

Bachelor of Science in Electrical and Computer Engineering, Shanghai Jiao Tong University 2020–2024

Selected Coursework: Abstract Algebra, Electromagnetism, Quantum Information Science, Quantum Mechanics (I-II), Advanced Computer Architecture, Compiler Construction, GPU Programming and Architecture, Operating Systems, Web Systems, Machine Learning

TECHNICAL SKILLS

C/C++, Python, Qiskit, PennyLane, CUDA, SystemVerilog, Rust, RISC-V, Julia, PyTorch, CPLEX Solver, CUDA-Q, Matlab, Slurm, React, Go, Stim, Qldpc, LLVM, Majorana simulator

PUBLICATIONS

TreeVQA: A Tree-Structured Execution Framework for Shot Reduction in Variational Quantum Algorithms (ASPLOS 2026) Yuewen Hou, Dhanvi Bharadwaj, Gokul Subramanian Ravi,

- Proposed a hierarchical execution framework to cluster VQA tasks for joint optimization and progressively branches into separate instances based on the Hamiltonian's dynamic landscape and similarity.
- Conducted a comprehensive evaluation showing that TreeVQA can reduce shot counts by 25.9x on average, and up to a maximum of 43.3x, over baseline VQA, for a fixed target application fidelity.

Dynamic Cloud Resource Management for Variational Quantum Algorithms (ASPLOS YArch 2024), Yuewen Hou, Gokul Subramanian Ravi,

- Enhanced quantum cloud computing fidelity and throughput through intelligent scheduling and resource estimation by leveraging stabilizer circuits.

QRIO: Quantum Resource Infrastructure Orchestrator *2024 IEEE International Symposium on Workload Characterization* (IISWC 2024), Shmeelok Chakraborty, Yuewen Hou, Ang Chen, and Gokul Subramanian Ravi,

- Designed an open-source distributed system automating quantum cloud resource access and allocation based on Kubernetes.
- Provided user-friendly quantum job specifications interface, including topology visualization, fidelity estimation, quantum gate error rates, and coherence time filtering.

SELECTED RESEARCH PROJECTS

Syndrome-Aware Noise Analysis and Compilation Optimization for Quantum Error Correction (Ongoing)

- Characterized the impact of coherent, non-Markovian noise on standard QEC codes (e.g., surface codes) using quasi-probabilistic decomposition-based stabilizer simulation.
- Developed ML-based syndrome analysis that incorporates hardware and circuit features to guide compilation and optimize syndrome measurement circuits.

Quantum-Classical Instruction Set Architecture Design (Ongoing)

- Exploring extensions to the RISC-V instruction set to enable high-performance hybrid programs with interleaved quantum and classical instructions.
- Taking a systems perspective on ISA design, balancing high-level quantum software requirements with implications for low-level quantum hardware architectures and controllers.

Operator Backpropagation Assisted Quantum State Preparation (Ongoing)

- Employed operator backpropagation in the Heisenberg picture and leveraged high-performance classical computing to efficiently simulate quantum ansätze to find good initial states for quantum applications with limited classical computational resources.

Multi-Start Clifford Initialization for the Quantum Approximate Optimization Algorithm (Under Review)

SELECTED ACADEMIC PROJECTS

Improving Qubit Movement Scalability of a Neutral Atom Array Quantum Computer Improved atom control architecture with delay-line memory to leverage the long coherence-time of neutral atom qubits and the ability to maintain entanglement while being physically separated in the spatial domain. Implemented a compression algorithm for the qubit shifting control waveform and decreased the waveform memory requirement to 1.4%.

Implementation of an R10K-style Out-of-Order CPU Led a group to build a fully synthesizable R10K-style N-way superscalar Out-of-Order CPU using SystemVerilog with advanced features including a non-blocking data cache, Gshare branch predictor, and multiport instruction cache.

HONORS AND AWARDS

NSF Software-Tailored Architecture for Quantum Co-Design Summer School Scholarship 2025

NSF Quantum Leap Challenge Institute for Robust Quantum Simulation Summer School Travel Grant 2025

Qiskit Global Summer School 2024 – Quantum Excellence 2024

University of Michigan CSE Department Fellowship 2024

University of Michigan Dean's List 2023

COMMUNITY SERVICE AND OPEN-SOURCE CONTRIBUTIONS

Journal Reviewer ACM Transactions on Quantum Computing

Program Committee for Artifact Evaluation MICRO-25, HPCA-26, ASPLOS-26

Unitary Hackers PauliPropagation.jl